IESTI05 – Edge AI

Machine Learning System Engineering

16.b Generative AI at the Edge

SLM (Small Language Models)

UNIFEI

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Professor Honoris Causa

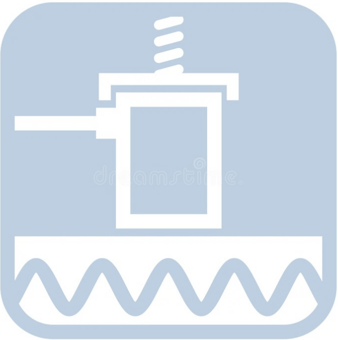
LLM / SLM

Large Language Model / Small Language Models

Perceptron (P)

Neural Network Architectures

Vibration Analysis



MLP - Deep Neural Network

Image Classification

CNN - Convolutional NN

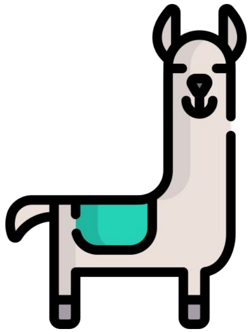
Text Generation

RNN - Recurrent NN (GRU/LSTM)

Image Generation

GAN - Generative Adversarial N.

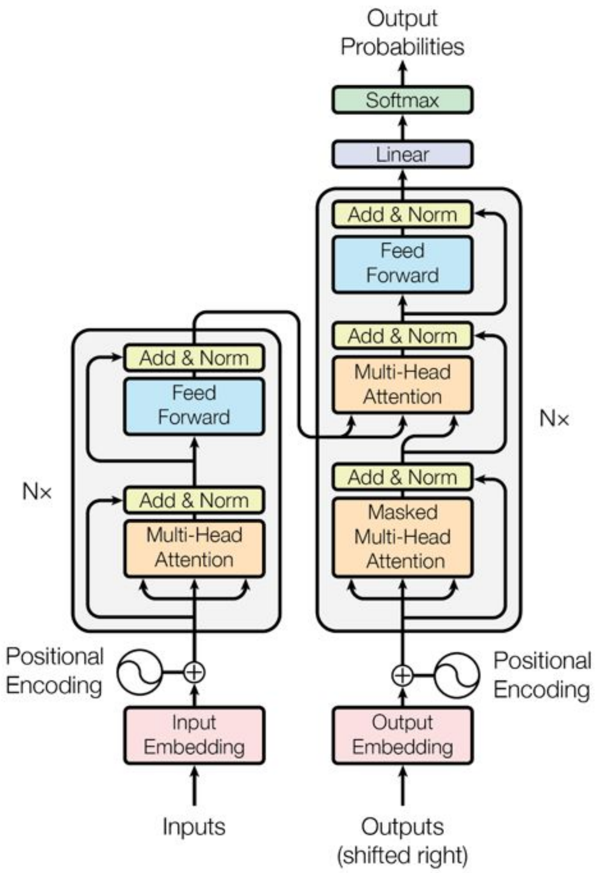
Large Language Models- LLMs



AN - Attention (Transformers)

LLM/SLM – Large /Small Language Model

Large Language Models (LLMs) and SLMs are advanced neural networks based on the Transformer architecture that excel in understanding and generating human language. They represent a significant evolution from earlier sequence-based mode like RNNs, which surpass them in handling long-range dependencies and parallel processing efficiency.



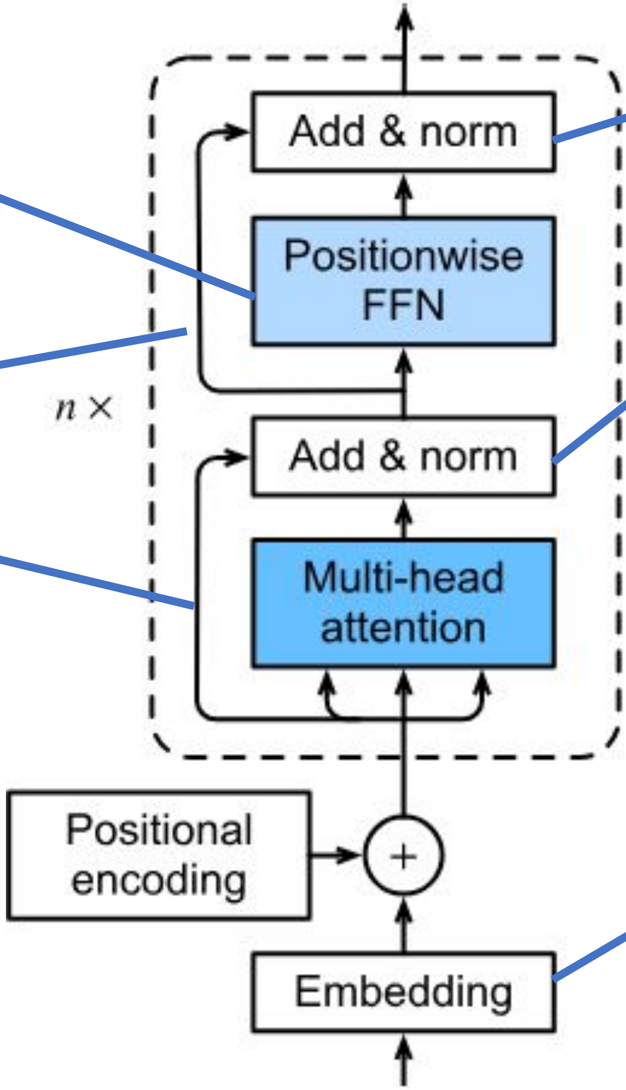
The Illustrated Transformer

Transformers

Dense Layer

Residuals Conexions
(CNN)

Normalization
(CNN)



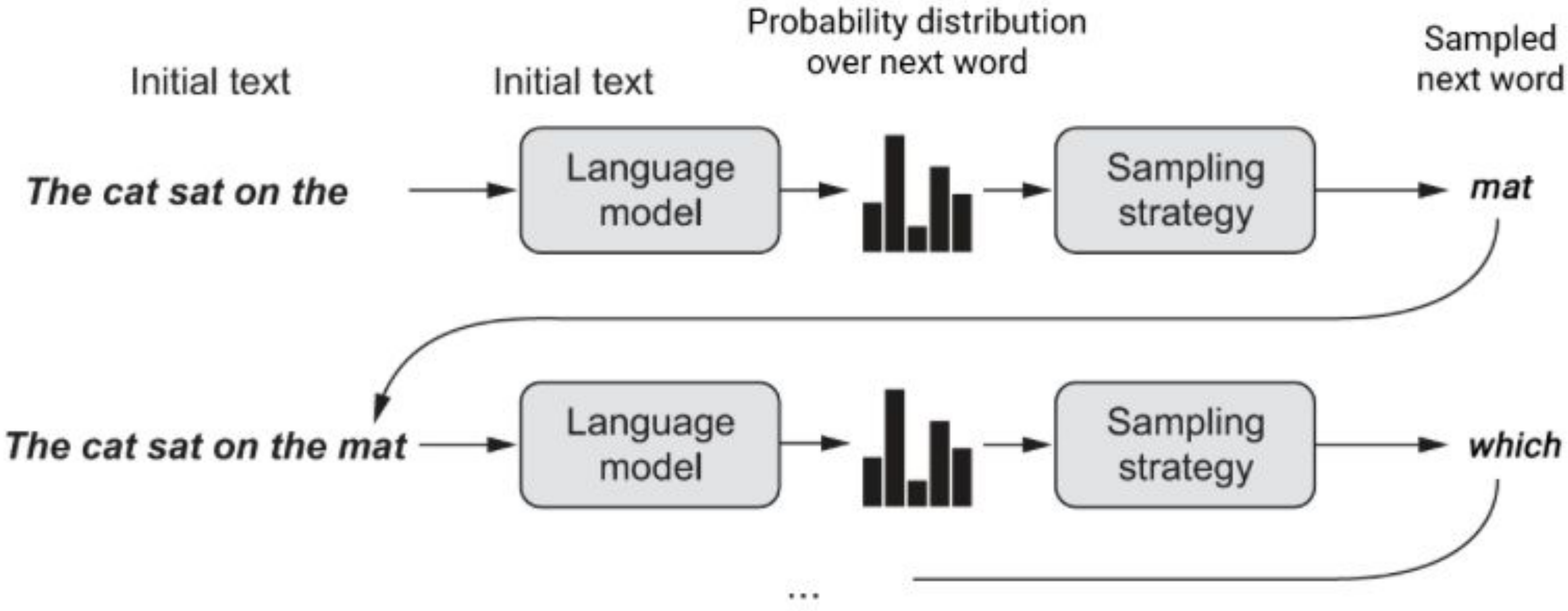
https://d2l.ai/_images/transformer.svg

NLP based on RNN

Transformer language
model visualization

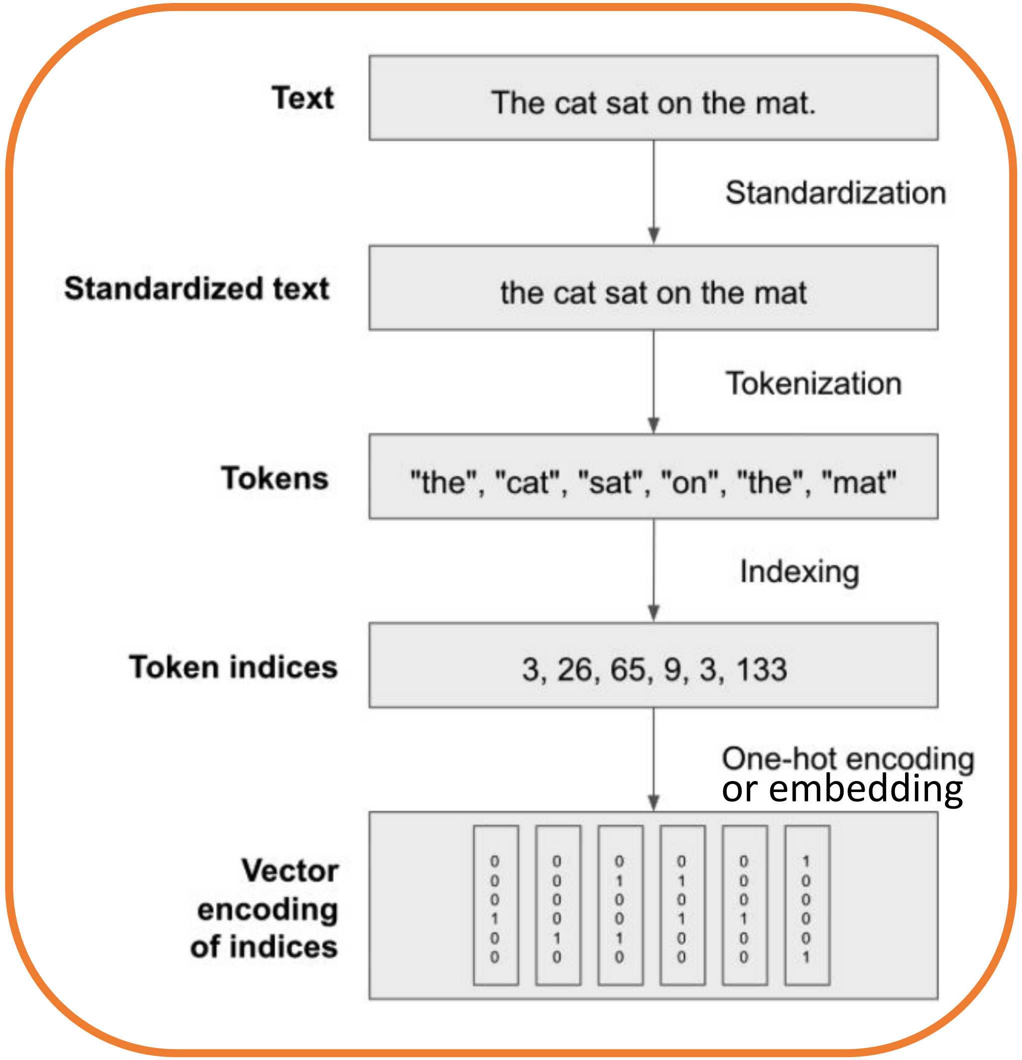
https://bbycroft.net/llm

LLM/SLM – Large /Small Language Model

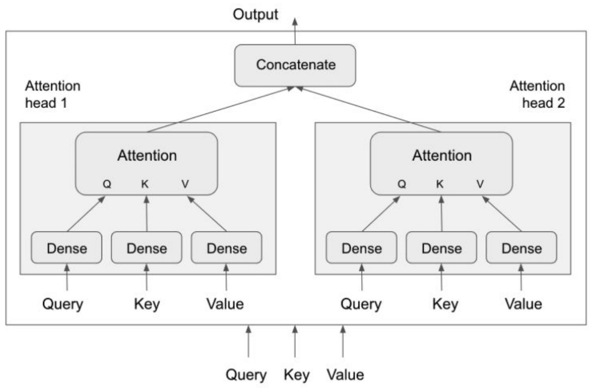


https://platform.openai.com/tokenizer

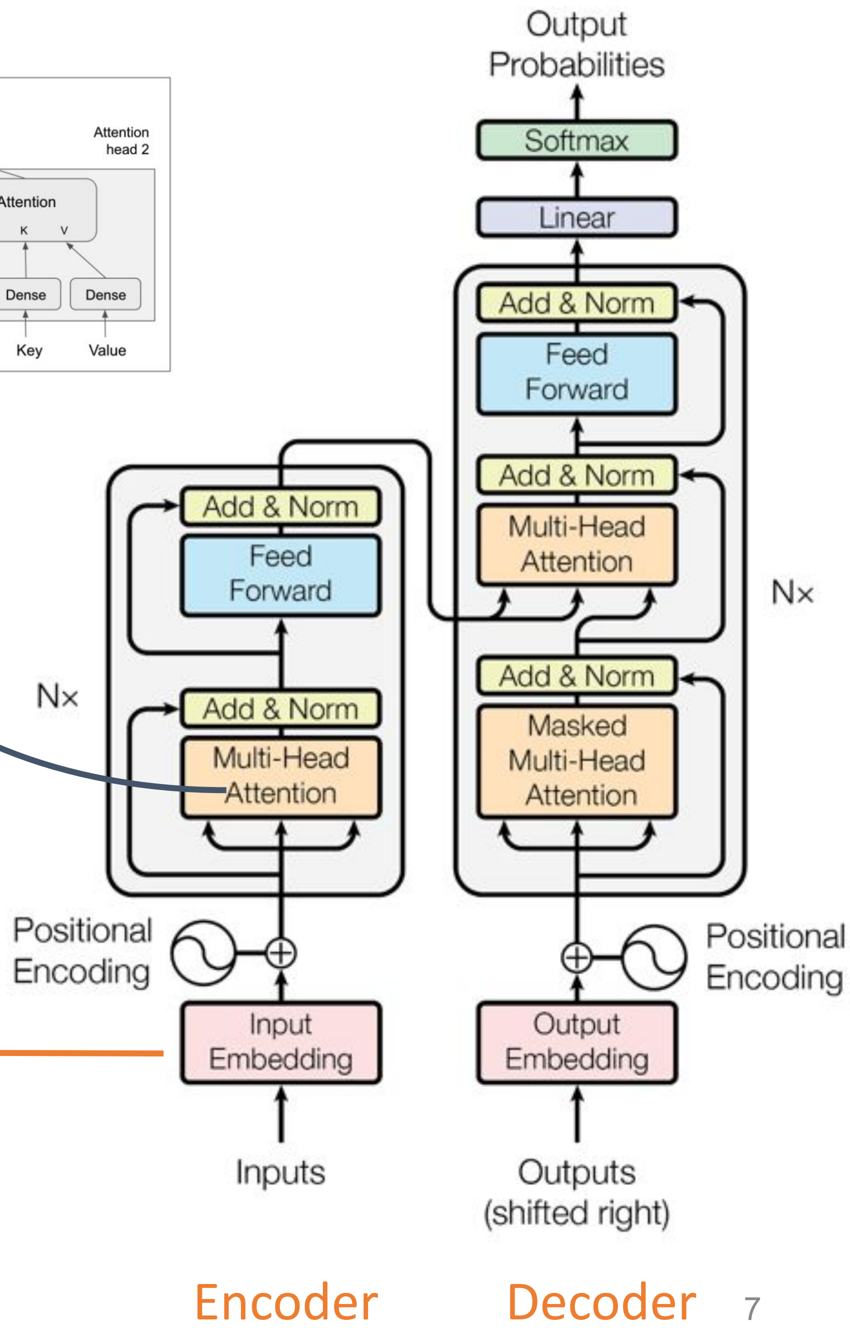
Embedding



Text
The cat sat on the mat.
Standardization
Standardized text
the cat sat on the mat
Tokenization
Tokens
"the", "cat", "sat", "on", "the", "mat"
Indexing
Token indices
3, 26, 65, 9, 3, 133
One-hot encoding
or embedding
Vector encoding of indices

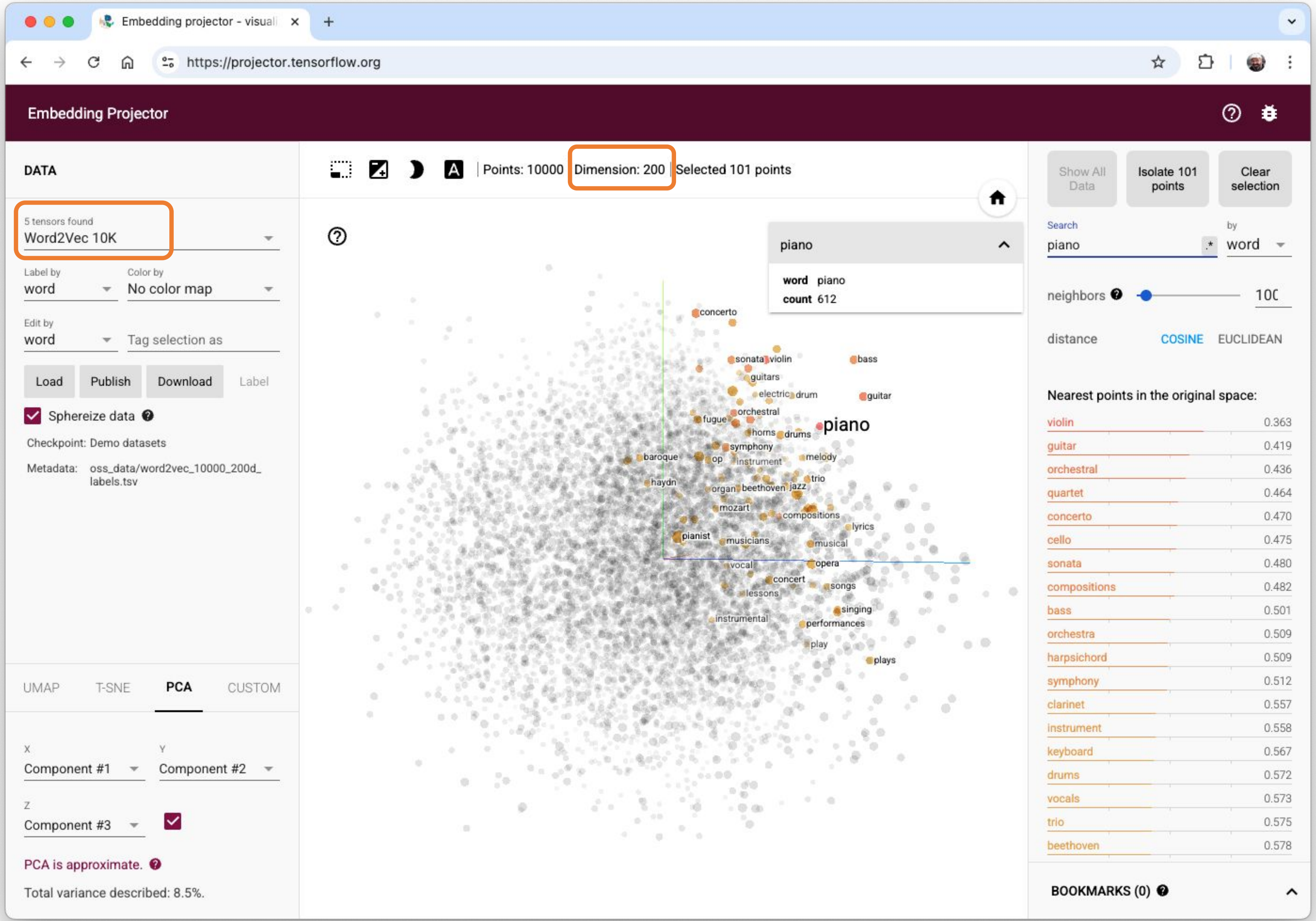


Output
Concatenate
Attention head 1
Attention head 2
Attention
Q K V
Dense
Dense
Dense
Query
Key
Value
Query
Key
Value

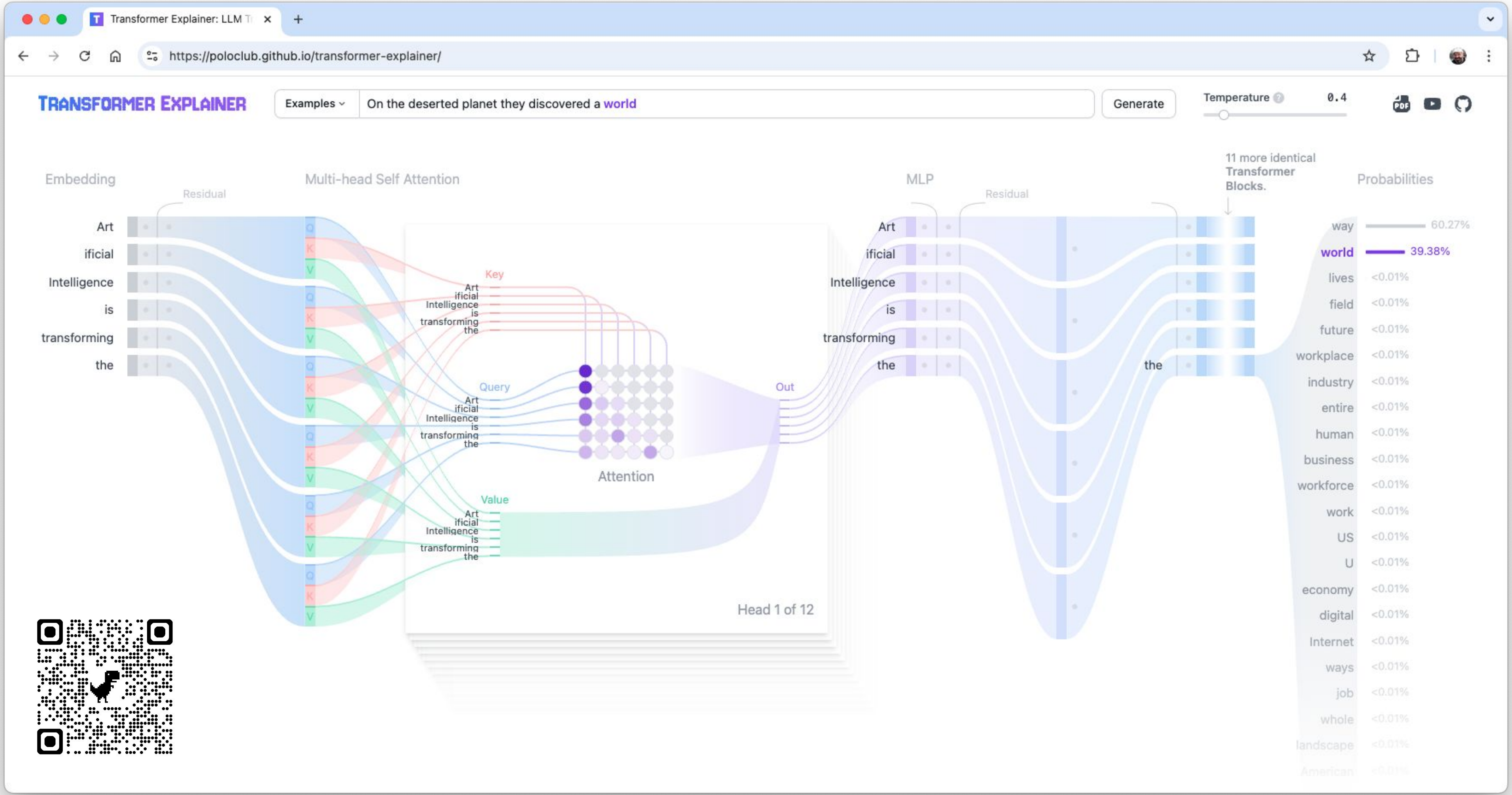


Output Probabilities
Softmax
Linear
Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention
Add & Norm
Masked Multi-Head Attention
Nx
Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention
Nx
Positional Encoding
Positional Encoding
Input Embedding
Output Embedding
Inputs
Outputs (shifted right)
Encoder
Decoder

Embedding



https://projector.tensorflow.org/



https://poloclub.github.io/transformer-explainer/

LLM TRAINING

Pretraining

Base model

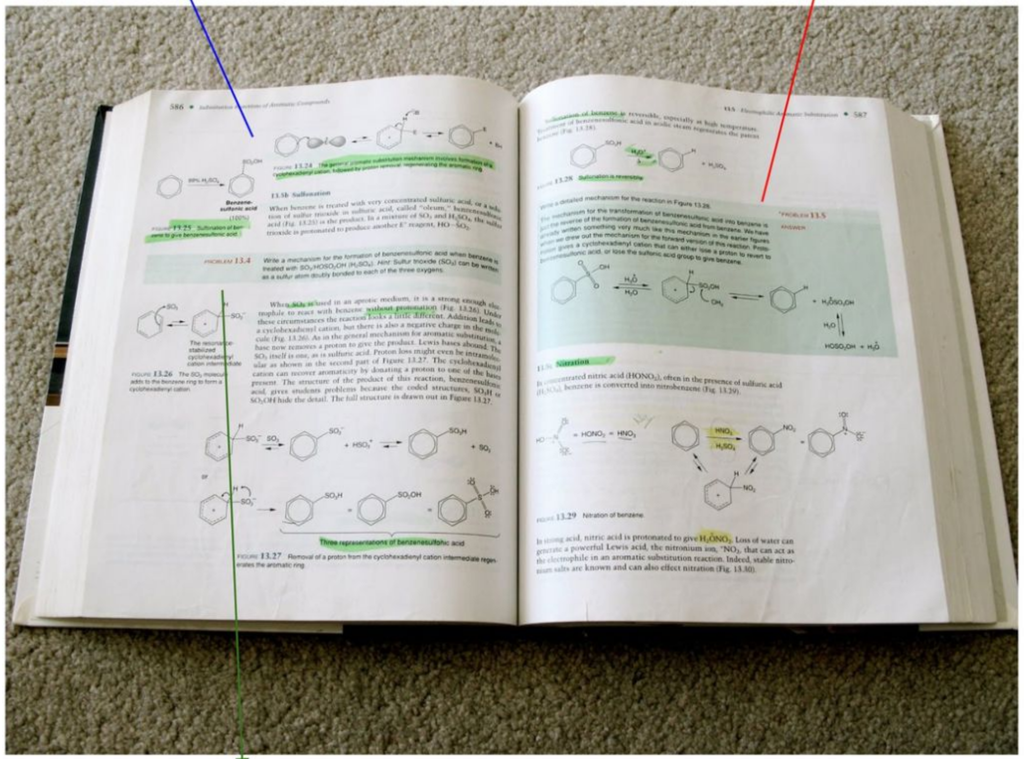
Post-Training
Supervised
Finetuning

SFT model

Post-Training
Reinforcement
Learning

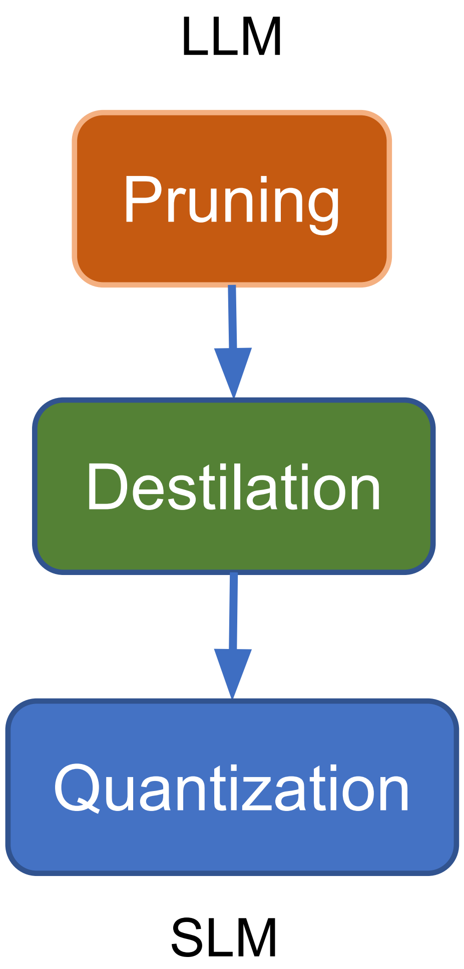
RL model

ChatGPT

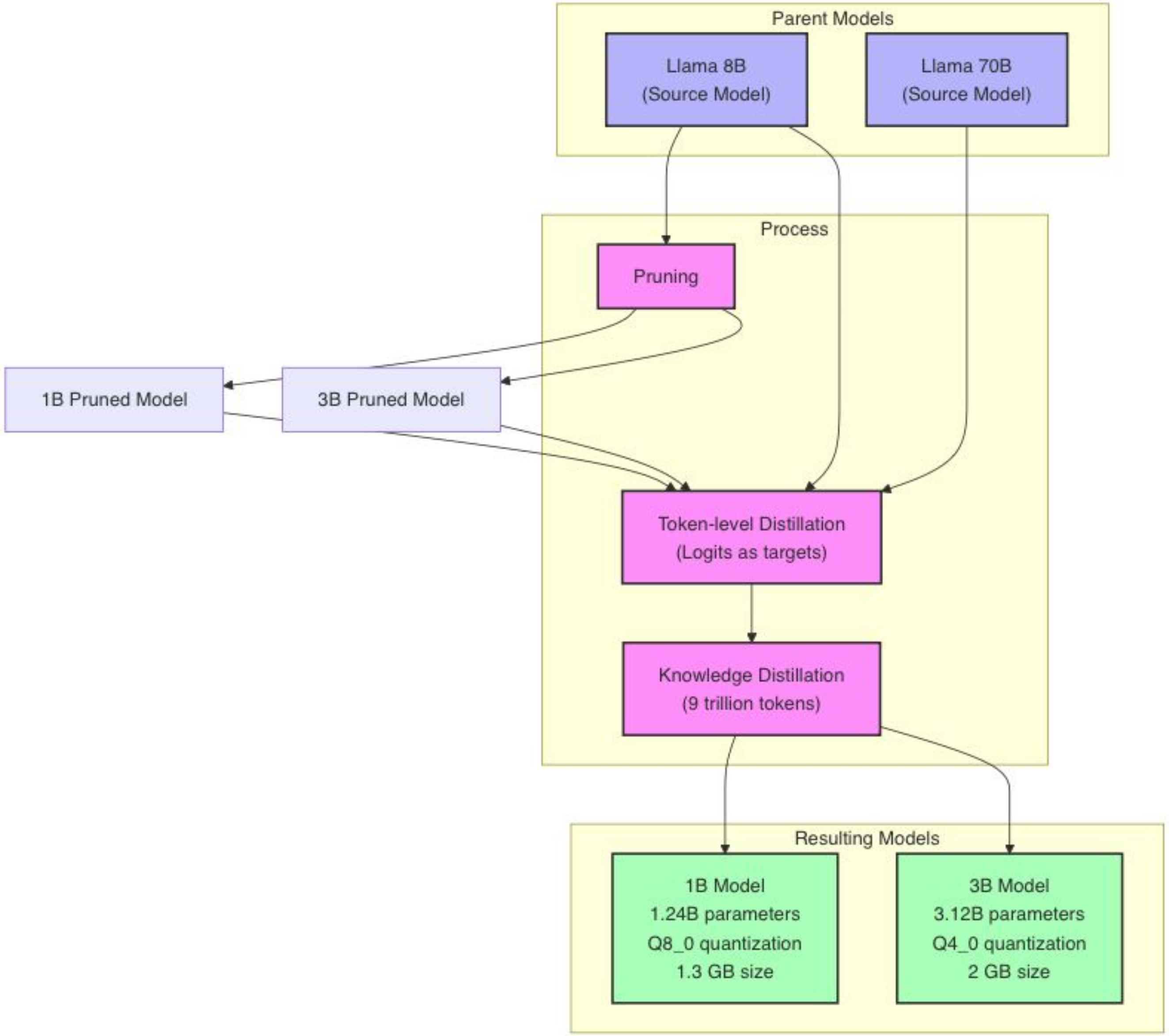


Deep Dive into LLMs like ChatGPT

Small Language Models

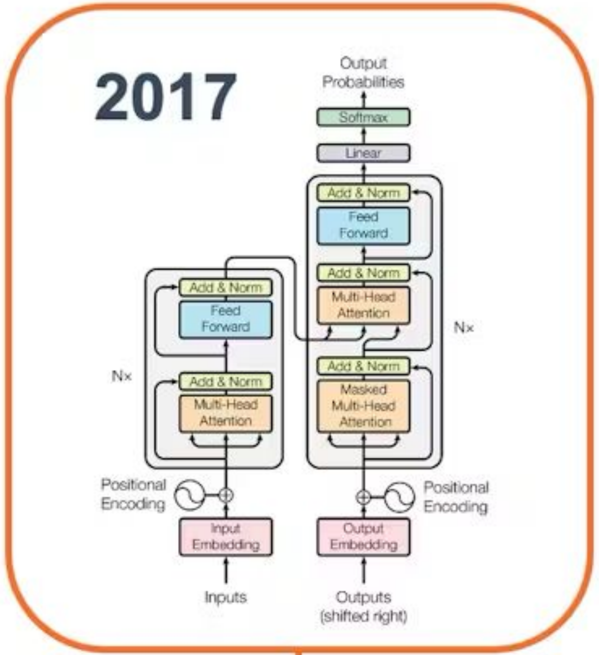


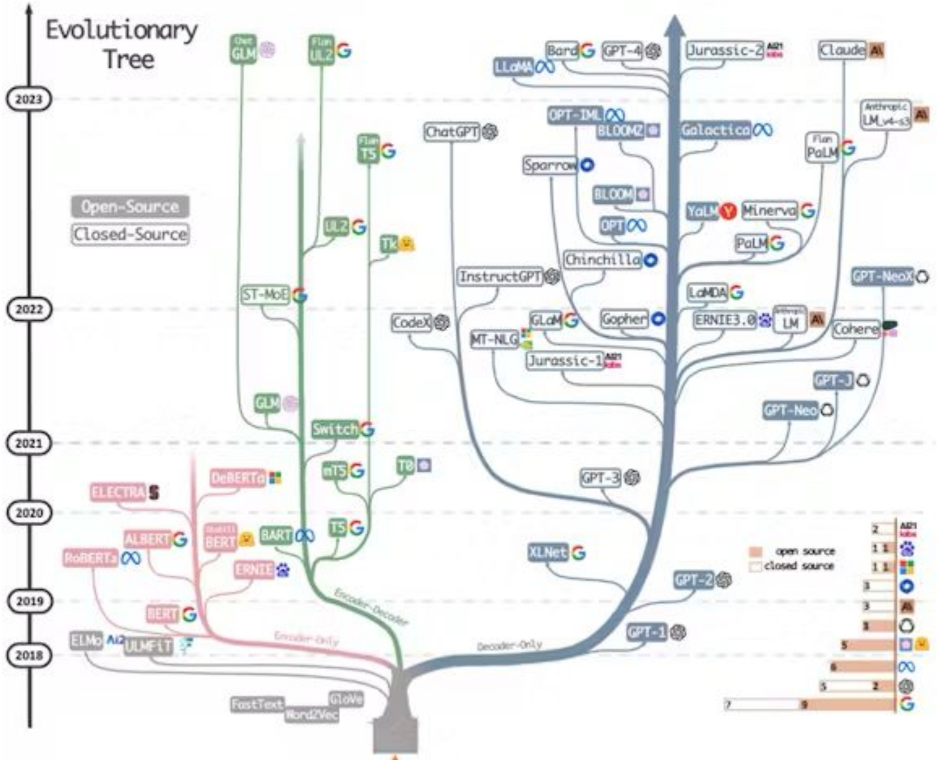
Llama 1B/3B

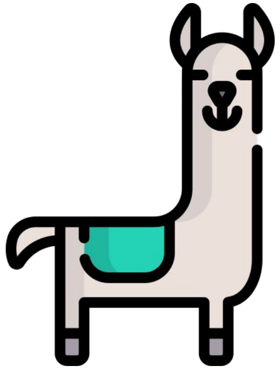


Transformers to LLMs and SLMs

	Open				Closed			
2025	Qwen	DeepSeek	Phi	Gemma	LlaMa	Gemini	GPT	Claude	Grok

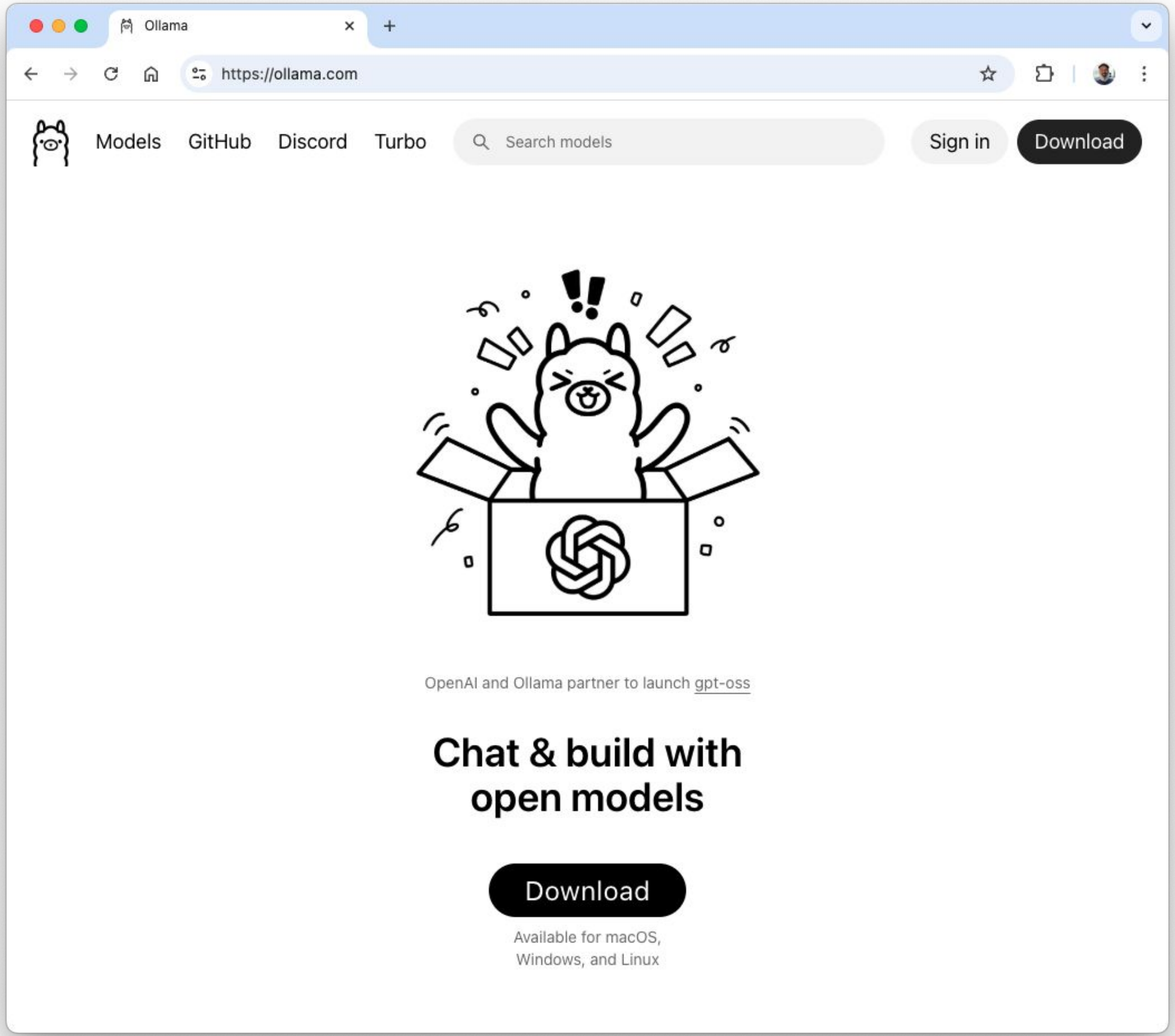






Ollama

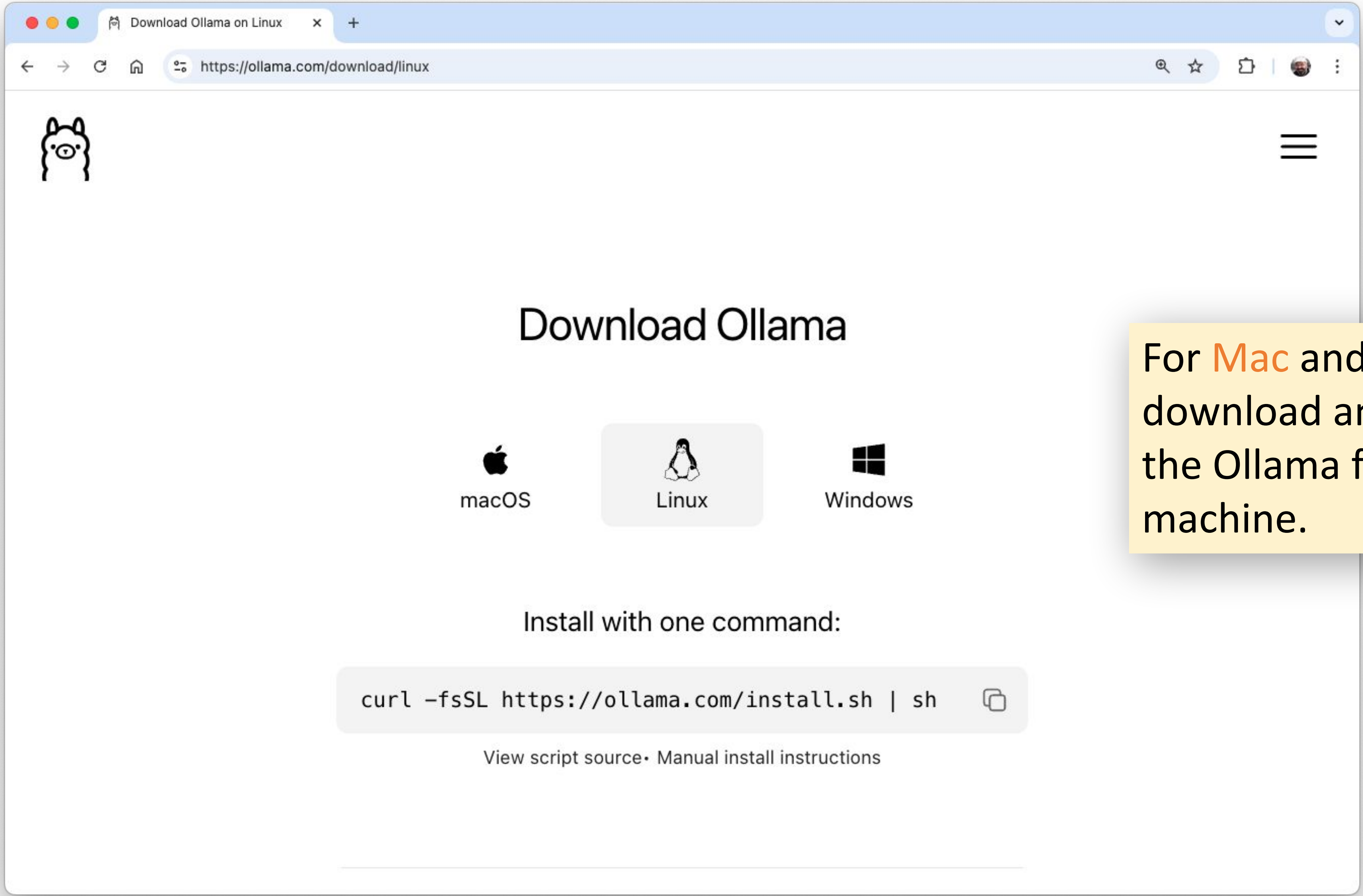
Running SLMs at the Edge



The primary and most user-friendly tool for running Small Language Models (SLMs) directly on a Raspberry Pi, especially the Pi 5, is Ollama. It is an open-source framework that allows you to install, manage, and run various SLMs (such as TinyLlama, smollm, Microsoft Phi, Google Gemma, Meta Llama, and LLaVa) locally on your Raspberry Pi for tasks like text generation and translation.

Alternatives options:

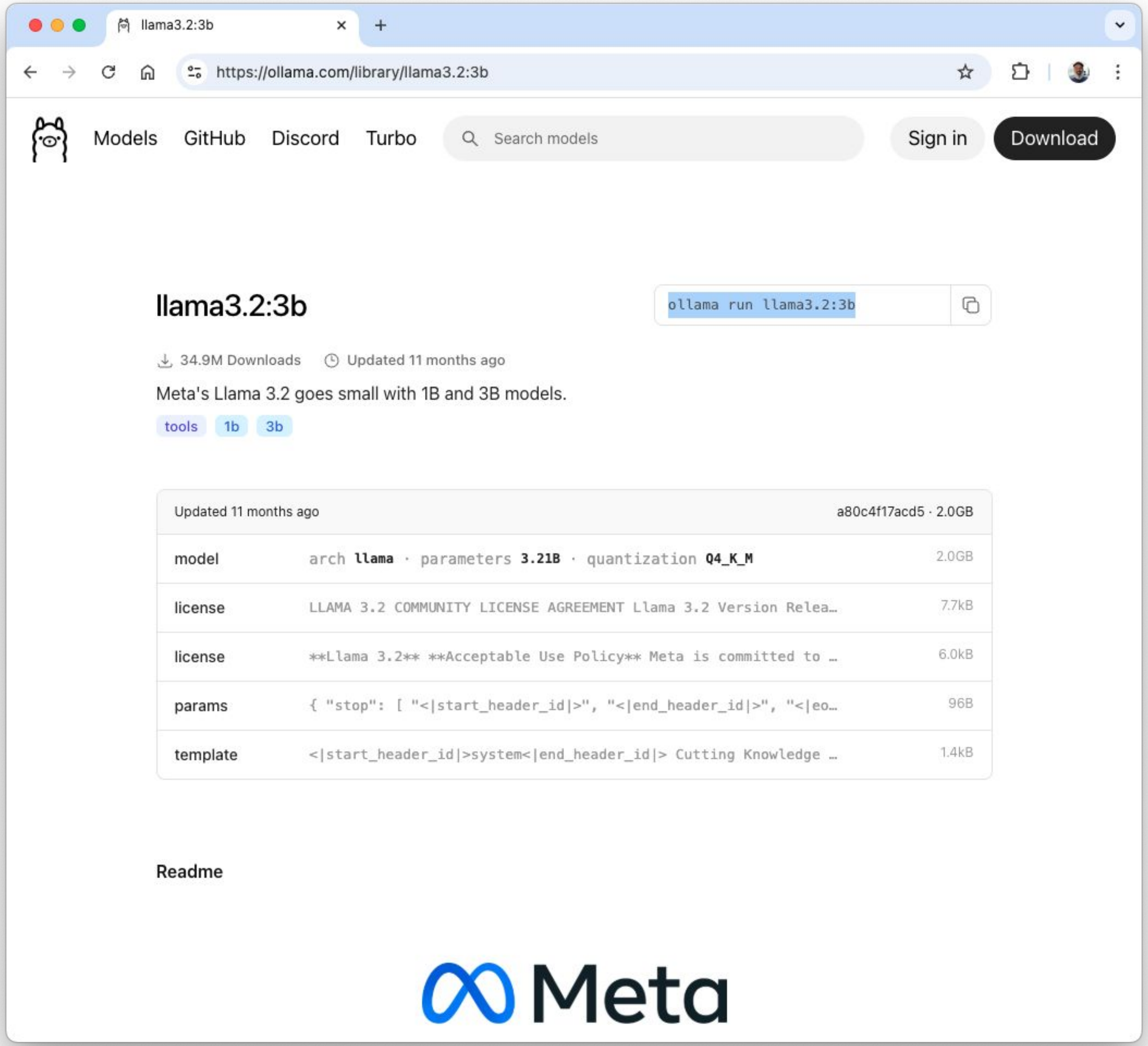
llama.cpp and the Hugging Face **Transformers** library are well-supported for running SLMs on Raspberry Pi. llama.cpp is particularly efficient for running quantized models natively, while Hugging Face Transformers offers a broader range of models and tasks, best suited for smaller architectures due to hardware limitations.

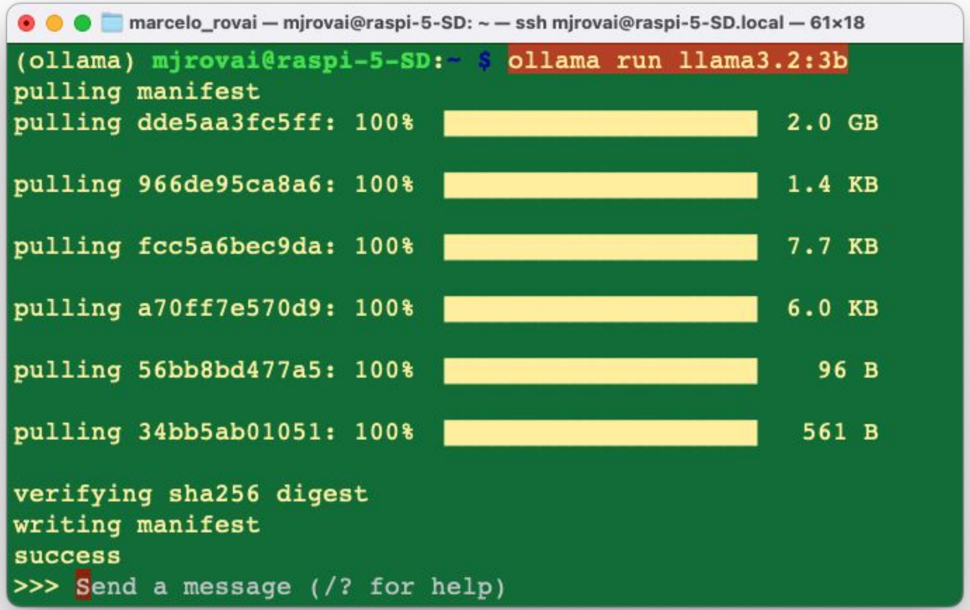


For Mac and Windows, download and execute the Ollama file on the machine.

marcelo_rovai — mjrovai@raspi-5-SD: ~ — ssh mjrovai@raspi-5-SD.local — 82×16

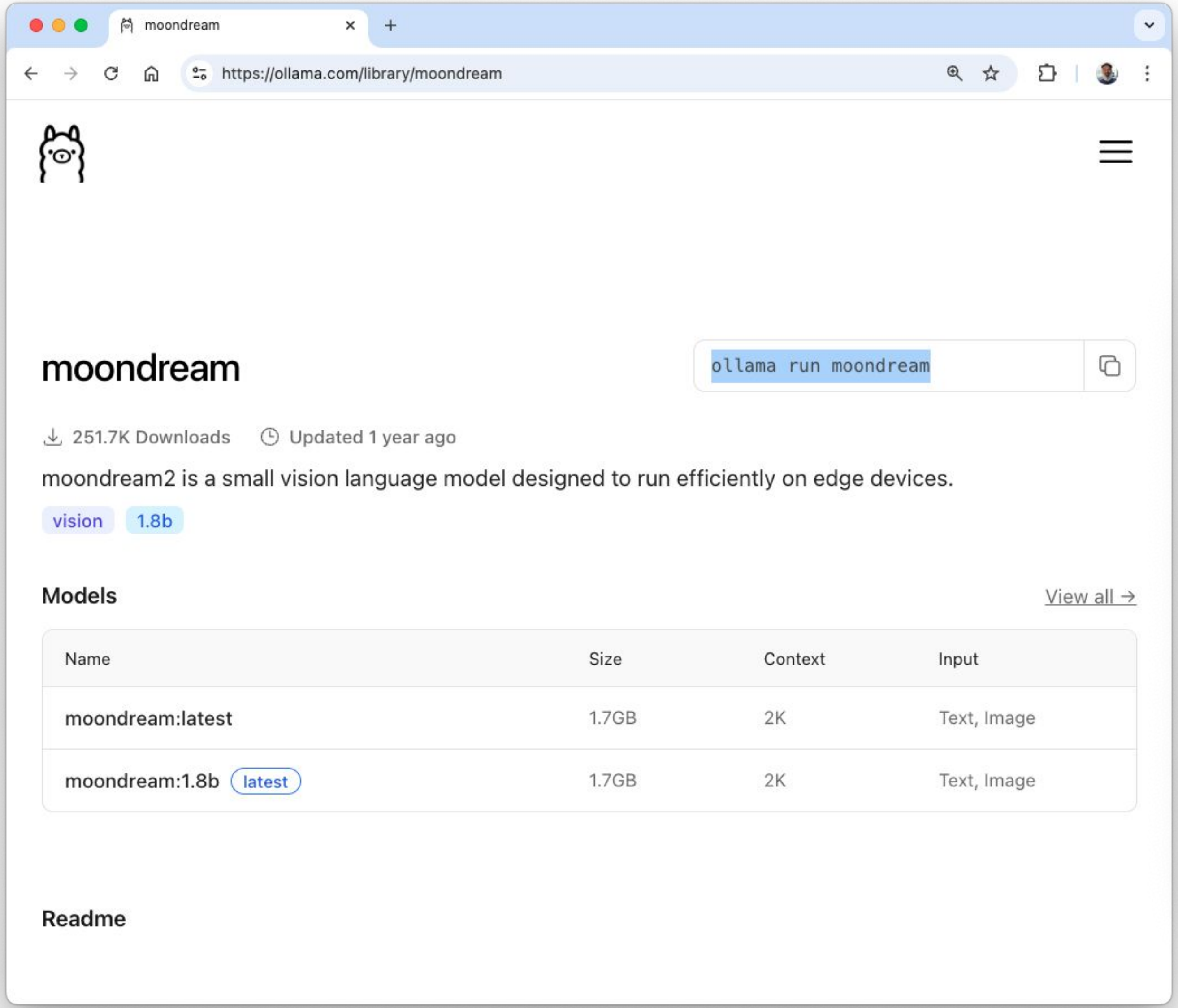
```
(ollama) mjrovai@raspi-5-SD:~ $ curl -fsSL https://ollama.com/install.sh | sh
>>> Installing ollama to /usr/local
>>> Downloading Linux arm64 bundle
######################################################################## 100.0%
>>> Creating ollama user...
>>> Adding ollama user to render group...
>>> Adding ollama user to video group...
>>> Adding current user to ollama group...
>>> Creating ollama systemd service...
>>> Enabling and starting ollama service...
Created symlink /etc/systemd/system/default.target.wants/ollama.service → /etc/sys
temd/system/ollama.service.
>>> The Ollama API is now available at 127.0.0.1:11434.
>>> Install complete. Run "ollama" from the command line.
WARNING: No NVIDIA/AMD GPU detected. Ollama will run in CPU-only mode.
(ollama) mjrovai@raspi-5-SD:~ $
```

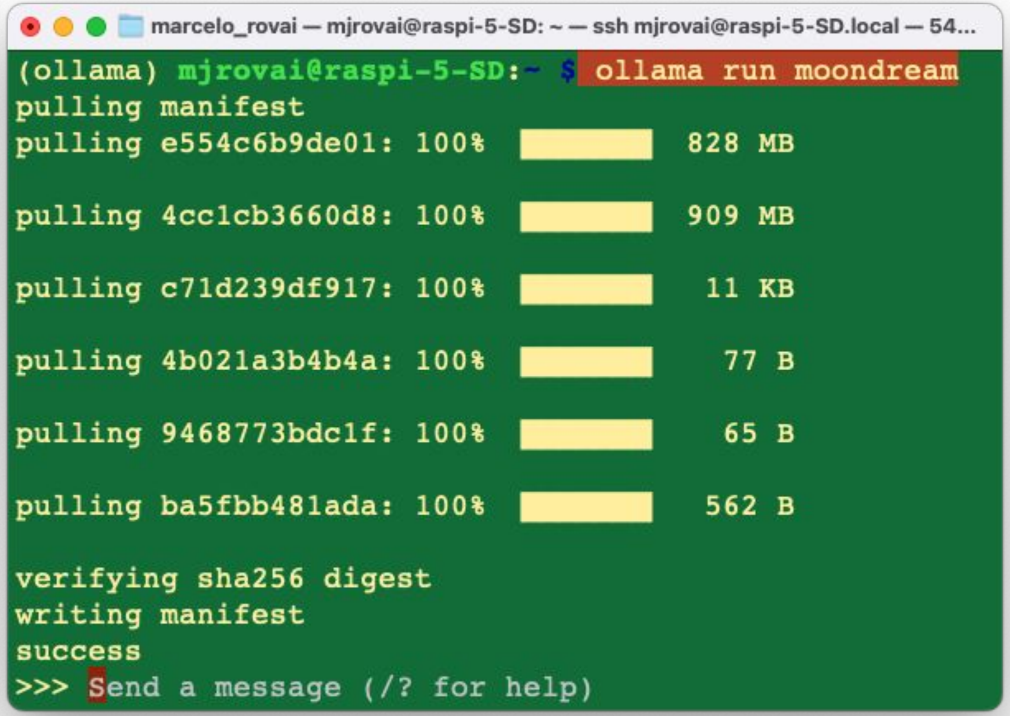

llama3.2:3b
https://ollama.com/library/llama3.2:3b
Models
GitHub
Discord
Turbo
Search models
Sign in
Download
llama3.2:3b
ollama run llama3.2:3b
34.9M Downloads
Updated 11 months ago
Meta's Llama 3.2 goes small with 1B and 3B models.
tools
1b
3b
Updated 11 months ago
a80c4f17acd5 · 2.0GB
model
arch llama · parameters 3.21B · quantization Q4_K_M
2.0GB
license
LLAMA 3.2 COMMUNITY LICENSE AGREEMENT Llama 3.2 Version Relea…
7.7kB
license
Llama 3.2 **Acceptable Use Policy** Meta is committed to …
6.0kB
params
{ "stop": ["<|start_header_id|>", "<|end_header_id|>", "<|eo…
96B
template
<|start_header_id|>system<|end_header_id|> Cutting Knowledge …
1.4kB
Readme
Meta


marcelo_rovai — mjrovai@raspi-5-SD: ~ — ssh mjrovai@raspi-5-SD.local — 61×18
(ollama) mjrovai@raspi-5-SD:~ $ ollama run llama3.2:3b
pulling manifest
pulling dde5aa3fc5ff: 100% 2.0 GB
pulling 966de95ca8a6: 100% 1.4 KB
pulling fcc5a6bec9da: 100% 7.7 KB
pulling a70ff7e570d9: 100% 6.0 KB
pulling 56bb8bd477a5: 100% 96 B
pulling 34bb5ab01051: 100% 561 B
verifying sha256 digest
writing manifest
success
>>> Send a message (/? for help)

marcelo_rovai — mjrovai@raspi-5-SD: ~ — ssh mjrovai@raspi-5-SD.local — 69×13

```
(ollama) mjrovai@raspi-5-SD:~ $ ollama run llama3.2:3b --verbose
>>> Qual a capital do Brasil?
A capital do Brasil é Brasília.

total duration:       5.140576457s
load duration:        185.275522ms
prompt eval count:    31 token(s)
prompt eval duration: 3.35620978s
prompt eval rate:     9.24 tokens/s
eval count:           9 token(s)
eval duration:        1.598257553s
eval rate:            5.63 tokens/s
>>> Send a message (/? for help)
```


moondream
https://ollama.com/library/moondream
moondream
ollama run moondream
251.7K Downloads
Updated 1 year ago
moondream2 is a small vision language model designed to run efficiently on edge devices.
vision
1.8b
Models
View all →
Name
Size
Context
Input
moondream:latest
1.7GB
2K
Text, Image
moondream:1.8b
latest
1.7GB
2K
Text, Image
Readme

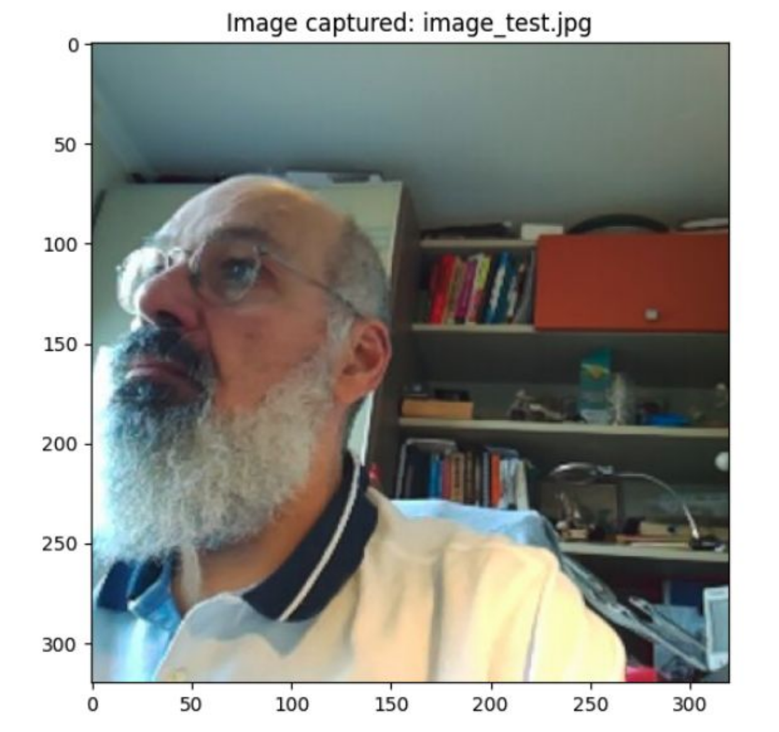

marcelo_rovai — mjrovai@raspi-5-SD: ~ — ssh mjrovai@raspi-5-SD.local — 54...
(ollama) mjrovai@raspi-5-SD:~ $ ollama run moondream
pulling manifest
pulling e554c6b9de01: 100% 828 MB
pulling 4cc1cb3660d8: 100% 909 MB
pulling c71d239df917: 100% 11 KB
pulling 4b021a3b4b4a: 100% 77 B
pulling 9468773bdc1f: 100% 65 B
pulling ba5fbb481ada: 100% 562 B
verifying sha256 digest
writing manifest
success
>>> Send a message (/? for help)

marcelo_rovai — mjrovai@raspi-5-SD: ~ — ssh mjrovai@raspi-5-SD.local — 87×18

```
(ollama) mjrovai@raspi-5-SD:~ $ ollama run moondream --verbose
>>> describe the image: /home/mjrovai/Documents/OLLAMA/image_test.jpg"
Added image '/home/mjrovai/Documents/OLLAMA/image_test.jpg'

The image features a man with glasses and a white beard, who is wearing a
collared shirt. He appears to be looking off into the distance or at something
out of frame. The room has several bookshelves filled with numerous books,
indicating that he might have an interest in reading or studying.

total duration:       54.36227837s
load duration:        63.257535ms
prompt eval count:    743 token(s)
prompt eval duration: 48.361595541s
prompt eval rate:     15.36 tokens/s
eval count:           62 token(s)
eval duration:        5.935988132s
eval rate:            10.44 tokens/s
>>> Send a message (/? for help)
```

marcelo_rovai — mjrovai@raspi-5-SD: ~ — ssh mjrovai@raspi-5-SD.loca...

```
>>> what is the color of the shirt?

The man's shirt is white.

total duration:        1m15.308297242s
```



Command	Description	Example Usage
ollama serve	Starts the Ollama server for local API access and inference	ollama serve
ollama run <model>	Runs a specified model interactively or with input text	ollama run smollm2:135m
ollama pull <model>	Downloads a model from the Ollama registry	ollama pull phi
ollama list	Lists all models currently downloaded and available	ollama list
ollama show <model>	Displays detailed information about a specific model	ollama show phi
ollama create <new_model> -f <file>	Creates a new model from a Modelfile (customization)	ollama create mymodel -f Modelfile
ollama rm <model>	Removes a specified model from your system	ollama rm phi

Command	Description	Example Usage
ollama ps	Shows currently running Ollama processes or sessions	ollama ps
ollama stop <model>	Stops a running model session or process	ollama stop phi
ollama cp <source> <target>	Copies an existing model to a new name	ollama cp phi mybackup
ollama push <model>	Uploads a model to the Ollama registry (requires account & API key)	ollama push mymodel
ollama --help	Lists all available commands and their descriptions	ollama --help
ollama --version	Shows the current version of Ollama installed	ollama --version
ollama <model> --verbose	Run the model, showing statistics at the end	ollama run phi --verbose

marcelo_rovai — mjrovai@raspi-5-SD: ~ — ssh mjrovai@raspi-5-SD.local — 77×5

```
>>> /bye
(ollama) mjrovai@raspi-5-SD:~ $ ollama list
NAME                ID              SIZE      MODIFIED
moondream:latest    55fc3abd3867    1.7 GB    10 minutes ago
llama3.2:3b         a80c4f17acd5    2.0 GB    21 minutes ago
```

marcelo_rovai — mjrovai@raspi-5-SD: ~ — ssh mjrovai@raspi-5-SD.local — 57×23

```
(ollama) mjrovai@raspi-5-SD:~ $ ollama show llama3.2:3b
  Model
    architecture        llama
    parameters          3.2B
    context length      131072
    embedding length    3072
    quantization        Q4_K_M

  Capabilities
    completion
    tools

  Parameters
    stop    "<|start_header_id|>"
    stop    "<|end_header_id|>"
    stop    "<|eot_id|>"

  License
    LLAMA 3.2 COMMUNITY LICENSE AGREEMENT

    Llama 3.2 Version Release Date: September 25, 2024

    ...
```

marcelo_rovai — mjrovai@raspi-5-SD: ~ — ssh mjrovai@raspi-5-SD.local — 54...

```
(ollama) mjrovai@raspi-5-SD:~ $ ollama show moondream
  Model
    architecture        phi2
    parameters          1.4B
    context length      2048
    embedding length    2048
    quantization        Q4_0

  Capabilities
    completion
    vision

  Projector
    architecture        clip
    parameters          454.45M
    embedding length    1152
    dimensions          2048

  Parameters
    stop           "<|endoftext|>"
    stop           "Question:"
    temperature    0

  License
    Apache License
    Version 2.0, January 2004
    ...
```

Questions?

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